Anand Bhattad

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Research Interests

My research span at the intersection of Computer Vision, Computer Graphics and Computational Photography. My current research focus is on Image-based Rendering, Inverse Rendering, Differentiable Rendering, Photorealistic Image Manipulation and other related topics.

Education

University of Illinois, Urbana-Champaign (UIUC), USA

Ph.D., Computer Science — *Jan'19 - Present*
Advised by Prof. David Forsyth

M.S., Computer Science — *Aug'17 - Dec'18*
M.S., Civil Engineering — *Aug'15 - Aug'17*

National Institute of Technology Karnataka (NITK) Surathkal, India — *GPA: 8.81/10.0*

B.Tech., Civil Engineering — *July'11 - May'15*

Research Internships

Intel ISL, Santa Clara/Champaign (remote); *Research Intern* — *Feb'21 - Present*
With Stephan Richter and Vladlen Koltun
Project on object rendering.

NVIDIA ADLR, Santa Clara/Champaign (remote); *Research Intern* — *May'20 - Dec'20*
With Aysegul Dundar, Andrew Tao and Bryan Catanzaro
Project on single image textured 3D inference via differentiable rendering.

Fyusion Inc, San Francisco; *Research Intern* — *May'19 - Aug'19*
With Abhishek Kar and Rodrigo Ortiz Cayon
Project on image-based relighting of cars.

Siemens Corporate Technology, Princeton; *Computer Vision & DL Intern* — *May'17 - Aug'17*
With Jan Ernst and Ziyan Wu
Developed a semi-automated segmentation tool based on Polygon-RNN from CVPR 2017.

Computational Sensor Lab, Michigan State University; *Research Intern* — *May'14 - July'14*
With Prof. Nizar Lajnef and Prof. Amir H. Alavi
Built a framework for Damage Detection Using Self-Powered Wireless Sensors

Earthquake Engineering Research Center, IIIT Hyderabad; *Research Intern* — *Jun'13 - July'13*
With Prof. Ramacharla Pradeep Kumar
Designed Earthquake Resistant Multi-Storied Building

Conference Papers

6. **View Generalization for Single Image Textured 3D Models**
A. Bhattad, A. Dundar, G. Liu, A. Tao, B. Catanzaro
Computer Vision and Pattern Recognition (CVPR), 2021

5. **Cut-and-Paste Neural Rendering**
A. Bhattad, D. A. Forsyth
Under review

4. **Unrestricted Adversarial Perturbations via Semantic Manipulation**
A. Bhattad*, M. J. Chong*, K. Liang, B. Li, D. A. Forsyth (* for equal contribution)
International Conference on Learning Representations (ICLR) 2020

3. **Improved Style Transfer with Calibrated Metrics**
M. Yeh, S. Tang, **A. Bhattad**, C. Zou, D. A. Forsyth
Winter Conference on Applications of Computer Vision (WACV) 2020

Workshop Papers

2. **Big but Imperceptible Adversarial Perturbations via Semantic Manipulation**
   A. Bhattad*, M. J. Chong*, K. Liang, B. Li, D. A. Forsyth (* for equal contribution)
   Computer Vision and Pattern Recognition (CVPR) Workshop 2019, ***Oral Presentation***

1. **Detecting Anomalous Faces with "No Peeking" Autoencoders**
   A. Bhattad, J. Rock, D. A. Forsyth
   Computer Vision and Pattern Recognition (CVPR) Workshop, 2018, ***Oral Presentation***

Teaching Experience

Computer Science, UIUC; *Graduate Teaching Assistant* *Jan'16 - Present*

- Applied Machine Learning (CS498 AML) in Fall'18 | Designed Kaggle competitions
- Data Structures (CS 225) in Spring'17 | Developed Assignments, Exams & Labs
- Introduction to Computing (CS 101) in Spring'16 & Fall'16 | **Rated as Excellent TA**

Other Projects

Impact of Graph Convolutional Network (GCN) for Active Learning *Mar'18 - May'18*
Project Mentor: Prof. ChengXiang Zhai `[Video | Report | Code]`

Pedestrian Detection and Automated Braking for Self-Driving Cars *Jan'17 - May'17*
Project Mentor: Prof. David Forsyth `[Video | Code]`

Services & Leadership

Organizer, External Speaker Series and Vision Lunch Jan 2021-Present

- Hosting the first external speaker series, inviting researchers outside of UIUC for technical talks
- Organizing computer vision reading and presentation group in the CS department of UIUC

Reviewer, Computer Vision Conferences Nov 2018-Present

- Served as reviewer for CVPR, NeurIPS, ECCV and ICCV conferences and workshops.

Program Committee Jun 2019

- Adversarial Machine Learning in Real-World Computer Vision Systems, CVPR 2019 Workshop
- Security and Privacy in Machine Learning, ICML 2019 Workshop

Organizer, Vision Coffee, Urbana-Champaign *May'18 - Aug'18*

- Vision and Deep Learning paper/topic weekly discussion group
- Finalized agenda/topic, location and initiated discussions

Co-Founder, SchoolEngg *Apr'13 - May'15*

- Forum for high-school students; providing the firsthand knowledge of all engineering disciplines
- Currently, renamed as *PrepLift*, an education counselling start-up in India

Co-Founder, American Society of Civil Engineers (ASCE) NITK Chapter *Apr'13 - May'15*

- Co-Founded the first ASCE student chapter of India
- Raised membership from 11 to 96 in a span of 6 months
- Coordinated with board members, faculty, and outside organizations to facilitate 10+ events

General Elections of India, Volunteer *Aug'13 - Apr'14*

- Helped register Voter Ids of 200+ people, from filling forms to getting their Voter IDs printed
- Spread awareness about the importance of voting through street-plays